# AI SKILL TREE 2026

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**Want to get better at AI, but don't know what you don't know? From daily chat (L1) to autonomous coding agents (L3–L4) and IMO-math / cyber-RL (L6), here is the full progression map.**

**① DIAGNOSE: Scan down each column (L1 → L6) to find where you stop**
**② LEVEL UP: Target the Top 3 pills in the row below**
**③ UNLEARN: Avoid the OBSOLETE traps on the right**

*Scope: Modern LLMs, Multimodal (mmLLM) & Agentic AI across end-users, MLE & model research (not GOFAI or classic ML/DL). Changes fast — see the AI Ladder Evolution mega-chart by Arian Prabowo · Updated: 2026-11-07 15:19:43 AEDT (UTC+11:00)

| AI LEVEL | THINK, SEARCH & ANALYZE | CREATE & DESIGN (MEDIA) | AUTOMATE & BUILD (CODE) | OBSOLETE |
|---|---|---|---|---|
| Overview · Persona · Surface · Use Case | Research · Data & Docs · RAG · Reasoning & Alignment | Image · Video · Audio · Voice · 3D · Provenance & DiT | Workflows · AI IDEs · Agents · Evals, Safety & Training | Still Doing This? You're Behind |
| **LEVEL 1** ▰▱▱▱▱▱ **EVERYDAY CONSUMER** | **ChatGPT / Claude Daily Q&A** | **ChatGPT Live Voice Mode** | **Claude Artifacts (Web App / SVG)** | **"Act as a..." Jailbreaks** |
| **Uses default consumer AI apps for conversational search, writing, media, and zero-install web mini-apps.**<br>• Persona: Casual daily user & general public.<br>• Surface: Phone app, web browser & built-in OS assistant on any phone or laptop.<br>• Use Case: Search replacement, drafting emails/texts, summarizing PDFs, live voice & browser mini-apps. | **Perplexity Cited Web Search** | **Conversational Image Gen (4o)** | **ChatGPT Canvas Live Preview** | **Buying 10k Prompt PDFs** |
| | **PDF & Article Upload Summary** | **Phone Camera Live Vision Q&A** | **Zero-Install Browser Vibe-Code** | **Blindly Trusting AI Claims** |
| | Google Search AI Overviews<br>Web Browsing in Chat<br>Apple AI Writing Tools<br>Hallucination Fact-Check | Gemini Mobile Assistant<br>Android Circle to Search<br>Meta AI in WhatsApp / IG<br>Draft & Rewrite Emails | Gemini Live Voice Chat<br>Mobile OS Voice Dictation<br>Canva Magic Media Studio<br>Screenshot OCR Translate | Google Lens Visual Q&A<br>Magic Eraser Photo Unblur<br>iOS Genmoji / Image Wand<br>Zoom / Meet AI Summaries | Single-File HTML/JS Mini-App<br>Gemini Canvas Web Preview<br>Browser Quiz & Flashcard App<br>1-Click Copy & Run Code | Interactive React Calculator<br>Shareable Artifact App Link<br>Interactive Chart & Dashboard<br>Fix Error by Pasting Message | ChatGPT w/o Web Search<br>ChatGPT Plugins Store<br>DALL-E mini / Craiyon<br>Passwords in Public Chat<br>AI as Medical/Legal Fact | Copy-Pasting 1 Para Only<br>Google Bard / Bing Sydney<br>Short 3-Word Prompts<br>AI Voice-Clone Scams<br>Ignoring Source Links |
| | DeepSeek Free Web / App<br>YouTube Video Summary<br>Travel Itinerary Planner<br>Memory On / Off Toggle | TLDR Thread Summarize<br>Gmail / Outlook Auto-Draft<br>Excel / Sheets Formulas<br>Data Training Opt-Out | Google Imagen / ImageFX<br>Spotify AI Playlist DJ<br>Voice Memo Transcript<br>Menu Photo Translation | Bing Image Creator<br>1-Click BG Removal<br>Google Photos Ask-Photos<br>Spot AI / Deepfake Tells | Grammar & Tone Polish<br>Foreign Language Chat<br>Click Cited Source Links<br>Temporary Incognito Chat | Auto-Captions on Video<br>Receipt Photo Expense<br>Whiteboard Photo Read<br>Short-Video Auto-Clip | Mermaid Flowchart in Chat<br>Split-Screen Code Preview<br>Interactive Budget Tracker<br>Download Single .html File | SVG Diagram & Logo Code<br>Ask AI to Explain Code<br>Landing Page Mockup in Chat<br>Iterative 'Make Button Blue' | Browser Mini-Game (Snake/2048)<br>Convert Table to Web App<br>Client-Side Only (No Keys)<br>Sandbox Preview Reset | |
| **LEVEL 2** ▰▰▱▱▱▱ **WORKSPACE POWER USER** | **Multi-Step Deep Research** | **Midjourney --sref / --cref Refs** | **Lovable / Bolt / v0 Web Apps** | **"Let's think step by step"** |
| **Orchestrates paid reasoning models, deep-research agents, and interactive canvases across multi-file workspaces.**<br>• Persona: Knowledge worker, analyst, designer & solo operator.<br>• Surface: Paid web workspaces, desktop apps, multi-source notebooks & creative studios.<br>• Use Case: Deep research reports, multi-doc synthesis, interactive canvases & consistent media. | **Extended Thinking (o3 / Claude)** | **Sora 2 / Veo 2 / Runway Video** | **ChatGPT Agent / Operator Mode** | **"Prompt Engineer" ($300k)** |
| | **Data Analysis & Chart Canvas** | **ElevenLabs Instant Voice Clone** | **Zapier / Make No-Code AI Flows** | **Uploading Client PII to Chat** |
| | Projects & Custom GPTs<br>DeepSeek-R1 DeepThink<br>Perplexity Pro Spaces<br>Zero-Data-Retention (ZDR) | NotebookLM 50-Source Pod<br>Gemini 1M–2M Token Upload<br>Multi-Doc Cross-Audit<br>Critique-Then-Revise Loop | NotebookLM Audio Host<br>Ideogram / 4o Text-Image<br>HeyGen Video Translate<br>Adobe Firefly Gen-Fill | Kling / Luma Keyframes<br>Suno v4 / Udio Song Gen<br>Superwhisper / Wispr Flow<br>Live Screen-Share Vision | Replit Agent / v0 Templates<br>Notion AI & M365 Copilot<br>Granola / Fireflies Notes<br>Custom GPT Action Webhooks | Scheduled AI Tasks & Alerts<br>Raycast AI Quick Commands<br>Airtable / Sheets AI Agents<br>Workspace SSO & Audit Logs | Tipping $200 / Threats<br>Midjourney Discord Bot<br>Manual 4k Text Splits<br>Copy-Pasting CSV Tables<br>Fake Legal Citations | "6 fingers" Neg Prompts<br>Custom GPT Store Hype<br>1-Page Prompt Walls<br>Manual Zoom Note-Taking<br>Shadow AI w/o IT SSO |
| | Grok-3 DeepSearch Mode<br>Context Token Trimming<br>Full-Book PDF Q&A<br>Rubric Self-Grading | LMSYS Chatbot Arena<br>CSV / Excel Code Sandbox<br>Earnings Call Synthesis<br>Role-Separated Debate | Pika Scene Ingredients<br>Captions.ai Avatar<br>Recraft Brand Vectors<br>Photoshop Gen-Expand | Descript Overdub & Eye<br>Hedra Character-2 Sync<br>Spline AI 3D Prompts<br>Commercial IP-Safe Models | Apple Shortcuts AI Actions<br>Superhuman / Gmail AI Rules<br>Tally / Typeform AI Routing<br>Readwise Reader AI Ghost | Browser AI Sidebar Actions<br>Slack AI Channel Digests<br>Coda AI Doc Automations<br>Mermaid / Whimsical AI Flow | Linear / Jira AI Triage<br>Loom AI Auto-Workflows<br>TextExpander / Alfred Snips<br>Role-Based Access (RBAC) | |
| **LEVEL 3** ▰▰▰▱▱▱ **VIBE CODER & BUILDER** | **Spec-Driven Dev (Spec Kit / PRD)** | **ComfyUI Ready Workflows** | **Cursor / Windsurf Agent Mode** | **AutoGPT / BabyAGI Loops** |
| **Ships full-stack apps and multi-step automations by directing AI IDEs, CLI coding agents, and cloud APIs.**<br>• Persona: Indie builder, automation maker & AI-native developer.<br>• Surface: AI IDEs (Cursor/Windsurf), CLI agents (Claude Code), app builders & APIs.<br>• Use Case: Natural-language app shipping, multi-file repo refactoring, MCP/webhook flows & API media pipes. | **Tavily / Exa / Firecrawl API** | **Fal.ai / Replicate Webhooks** | **Claude Code CLI & CLAUDE.md** | **Agent Deleting Failing Tests** |
| | **Repomix / Gitingest Packing** | **ElevenLabs / Vapi Voice API** | **SKILL.md & Subagent Workflows** | **Blindly Running Agent rm -rf** |
| | System vs User Role Split<br>Jina Reader URL-Markdown<br>Markdown Memory Banks<br>Few-Shot JSON Examples | llms.txt Docs Standard<br>Context Window Budgeting<br>Multi-Step Prompt Chains<br>OpenRouter Benchmarking | Black Forest Flux API<br>Whisper + FFmpeg Subs<br>Recraft v3 Vector API<br>Magnific / Krea Upscale | Remotion Code-to-Video<br>Sync.so Lip-Sync API<br>Figma-to-Code Plugins<br>Luma / Meshy 3D Assets | .cursor/rules & AGENTS.md<br>Context7 Live Docs MCP<br>Human-in-Loop Tool Approve<br>Browser-Use / Stagehand | Claude Desktop MCP & n8n<br>Plan-Before-Act & Diffs<br>Codex CLI / OpenCode / Aider<br>Vercel AI SDK & Supabase | API Keys in Public Git<br>Legacy .cursorrules Dump<br>Manual Regex Scraping<br>No Git Commit Pre-Agent<br>No API Spend / Rate Cap | Poisoned Rules / Issue Text<br>50-Node Flowise Chains<br>Pasting Files to Web UI<br>Shipping AI SQL w/o Auth<br>Fake Slopsquat Packages |
| | Perplexity Sonar API<br>Prompt Git Versioning<br>Streaming SSE Responses<br>Needle-in-Haystack Check | Brave Search Grounding<br>Token Cost & Spend Caps<br>Temp / Top-P API Tuning<br>Untrusted Input Delimiters | Runway Act-One API<br>AssemblyAI Diarization<br>LivePortrait Face API<br>Gladia Multilingual ASR | Hume Empathic Voice API<br>Creatomate JSON Video<br>Mux Video Auto-Chapters<br>C2PA / SynthID Metadata | Retell Phone Webhooks<br>Photoroom Batch API<br>PlayHT Low-Latency TTS<br>NSFW Moderation API Gate | Ollama / LM Studio Local<br>Together / Fireworks API<br>Red-Green TDD & E2E Tests<br>GitHub Actions PR Bot | Groq / Cerebras LPU API<br>Helicone / LangSmith Logs<br>Secret & API-Key Scrubbing<br>Slopsquat Package Check | LiteLLM / CF AI Gateway<br>OWASP AI Code & Rules Check<br>Regex / Bash / Git via AI<br>Stripe + Auth0 Vibe Ship | | |
| **LEVEL 4** ▰▰▰▰▱▱ **AI & AGENT ENGINEER** | **Pydantic / Zod Schema Outputs** | **OpenAI / Gemini WebRTC API** | **Custom MCP Servers & OAuth** | **Shipping w/o Evals or Traces** |
| **Architects production multi-agent backends with schema-enforced tool use, hybrid RAG, guardrails, audit traces, and CI evals.**<br>• Persona: Production AI software engineer & systems architect.<br>• Surface: Python/TS SDKs, custom MCP servers, state graphs, microVM sandboxes & observability stacks.<br>• Use Case: Stateful multi-agent backends, hybrid RAG, guardrails, audit trails, compliance & CI eval gates. | **Hybrid RAG + Cross-Rerankers** | **LiveKit / Pipecat Voice AI** | **LangGraph Durable State Graphs** | **Naive 500-Tok Vector RAG** |
| | **ColPali & Docling PDF RAG** | **Programmatic ComfyUI API** | **Promptfoo / Braintrust CI Evals** | **Regex-Parsing LLM JSON** |
| | Prompt Caching (90% Saved)<br>Anthropic Thinking Tokens<br>DSPy Prompt Optimization<br>Cohere Rerank-v3.5 Pipe | Hallucination & Faithfulness<br>Contextual & GraphRAG<br>BAML Typed Prompt Calls<br>Citation Span Grounding | Deepgram Nova-3 Stream<br>Vision-LLM UI Grounding<br>Twilio SIP Call Bridging<br>CLIP / SigLIP Visual RAG | Cartesia / Kokoro TTS<br>Barge-In VAD Interruption<br>TwelveLabs Video Search<br>WebSocket PCM16 Streams | E2B / Bubblewrap Sandboxes<br>Llama-Guard / NeMo Rails<br>Recursive Harness (Harness²)<br>Qdrant / Milvus / pgvector | Audit Trails & OTel Traces<br>LLM-as-a-Judge Rubrics<br>OpenAI SDK / PydanticAI<br>SWE-bench Eval Harness | Un-Sandboxed Code Exec<br>LangChain LLMChain Class<br>text-embedding-ada-002<br>Unicode Tag / ANSI Smuggling<br>No Model Fallback Route | Fine-Tuning to Add Facts<br>Assistants API v1 Polls<br>Pinecone Pod-Index Lock<br>No Tool Audit Logging<br>Vibe-Check Only Evals |
| | Hypothetical Embed (HyDE)<br>Corrective RAG (CRAG)<br>Surya / LayoutLM Tables<br>Faithfulness Gate Check | Parent-Doc Late Chunking<br>Semantic Intent Router<br>Demographic Fairness Eval<br>Time-Weighted Memory | Frame-Accurate Video Cut<br>HeyGen Streaming Avatar<br>Bounding-Box Detection<br>Sub-500ms Voice Loop | Silero VAD Turn-Taking<br>WhisperX Word Timestamps<br>Deepfake / Spoof Detection<br>Audio Watermark Check | Daily.co WebRTC Mesh<br>pyannote Speaker Diarize<br>C2PA Cryptographic Sign<br>3D Gaussian Asset API | PyRIT / Garak Red-Teaming<br>Human-in-Loop Interrupts<br>Redis Semantic Caching<br>Canary / Shadow Deploys | EU AI Act / SOC2 Audit Logs<br>Agent Memory (Mem0/Letta)<br>Langfuse / Phoenix Spans<br>Confused-Deputy & RBAC Gate | Harness Behavior-Code Audit<br>Temporal Durable Exec<br>Cascade Model Cost Router<br>A2A Agent2Agent Protocol | | |
| **LEVEL 5** ▰▰▰▰▰▱ **LOCAL & POST-TRAINING** | **Synthetic CoT Dataset Curation** | **ComfyUI Custom Node Pipelines** | **vLLM / SGLang GPU Serving** | **Full-Weight 7B Style FT** |
| **Serves quantized open-weight models on dedicated GPUs/Macs and aligns weights via QLoRA, DPO, GRPO, and steering.**<br>• Persona: ML systems engineer, local-first builder & model post-trainer.<br>• Surface: Multi-GPU rig (RTX 3090/4090/5090), unified-memory Mac, vLLM, SGLang & Unsloth.<br>• Use Case: High-throughput local serving, QLoRA/DPO/GRPO alignment, weight merging & air-gapped AI. | **Rejection-Sampled SFT / DPO** | **Flux.1 / Wan-2.1 Local Weights** | **Unsloth / Axolotl / TRL QLoRA** | **Automatic1111 SD 1.5 UI** |
| | **Grammar-Constrained GBNF / JSON** | **Kohya / AI-Toolkit LoRA Train** | **DPO / ORPO / Local GRPO Tune** | **SFT Without KL / Eval Checks** |
| | Local DeepSeek-R1 & Qwen-3<br>RoPE Scaling (YaRN)<br>KTO / IPO / SimPO Alignment<br>Air-Gapped Hybrid RAG | Constitutional / RLAIF Data<br>Min-P / DRY / XTC Sample<br>Custom ChatML Templates<br>Domain Eval Benchmarks | ControlNet & IP-Adapter<br>Kokoro-82M / F5-TTS Local<br>Florence-2 & SAM-2 Vision<br>TensorRT Diffusion Accel | Whisper-v3-turbo Local<br>RVC / So-VITS Voice Clone<br>GGUF FP8 Flux Offload<br>HunyuanVideo / Mochi Run | EXL2 / AWQ / GGUF / FP8<br>Outlines / XGrammar FSM<br>DoRA / rsLoRA / PiSSA / IA3<br>KV-Cache FP8 / INT4 Quant | SGLang RadixAttention<br>Speculative Decode (EAGLE)<br>Activation Steering Vectors<br>Apple MLX / Dual-4090 Rig | 8-bit bitsandbytes Slow<br>Manual GPTQ Calibration<br>LoRA Rank-128 Overfit<br>Static RoPE Alpha Hacks<br>Blind Frankenmerges | Oobabooga Text-Gen UI<br>DreamBooth 40m Colab<br>Raw Alpaca 52k SFT Set<br>Flask Single-Batch API<br>FT Without Safety Evals |
| | Refusal Boundary Tuning<br>KL-Divergence Drift Check<br>System Prompt Distill<br>CoT Trace Length Tuning | Bias & Toxicity Filtering<br>Continued Pretraining (CPT)<br>Local BGE-Reranker<br>RAGAS Local Faithfulness | TeaCache / WaveSpeed<br>AnimateDiff Motion Nodes<br>CosyVoice2 / Fish-Speech<br>JoyCaption VLM Tagging | Chatterbox / Orpheus TTS<br>Depth-Anything ControlNet<br>Ultimate-SD Tiled VAE<br>Concept Erasure LoRA | faster-whisper CTranslate<br>InstantID / FaceID Nodes<br>SageAttention Speedup<br>Watermark-Preserving Gen | llama.cpp / KoboldCpp<br>Qwen-Coder-32B Local IDE<br>Hermes-3 Local Tool Call<br>Prefill-Decode Disagg | MergeKit SLERP/TIES/DARE<br>Machine Unlearning Runs<br>OmniParser GUI Grounding<br>Zeroth-Order (MeZO) Opt | vLLM Multi-LoRA Serving<br>Local Stdio MCP & SQLite<br>GCG / AutoDAN / BoN Red-Team<br>imatrix Importance Quant | | |
| **LEVEL 6** ▰▰▰▰▰▰ **FRONTIER RESEARCHER** | **RLVR (Verifiable Reward RL)** | **Rectified Flow Matching (DiT)** | **FSDP2 / Megatron 3D Parallel** | **4-Model PPO RLHF Pipeline** |
| **Pre-trains foundation models, designs verifiable RL & scalable-oversight alignment, and authors custom GPU kernels.**<br>• Persona: Foundation model scientist, alignment researcher & kernel author.<br>• Surface: Multi-node H100/B200/TPU clusters, InfiniBand fabrics, FSDP2/Megatron & CUDA/Triton.<br>• Use Case: Base/MoE pre-training, RLVR/GRPO, mechanistic interpretability, scalable oversight & GPU kernels. | **GRPO (Group Relative Policy)** | **Native Early-Fusion Omni-Modal** | **Custom Triton & FlashAttention-3** | **Dense-Only 500B+ Pretrain** |
| | **Test-Time Compute Scaling** | **3D Spatio-Temporal Video VAE** | **DeepSeek MLA & Aux-Free MoE** | **Unverifiable Code/Math RMs** |
| | Scalable Oversight & Debate<br>Process Reward Model (PRM)<br>AlphaProof & OpenMath Lean<br>Pass@k → Pass@1 Distill | Reward Hacking Defense<br>MCTS Tree-Search Reason<br>Deliberative CoT Alignment<br>Benchmark Decontam LSH | Discrete Audio Tok (Mimi)<br>Autoregressive Visual Tok<br>Chameleon / Transfusion<br>Full-Duplex Speech Model | Consistency Flow Distill<br>Interleaved Omni Attend<br>3D Gaussian Splatting<br>JEPA / Genie-2 World Sim | Mech Interp & SAE Circuits<br>Exec-Guided Code RLVR<br>ThunderKittens / CUTLASS<br>InfiniBand & NCCL Tuning | FP8 / MXFP8 Mixed Train<br>XBOW / Cybench Zero-Day RL<br>torch.compile & Inductor<br>OpenRLHF / VeRL Clusters | Scaling w/o Safety Evals<br>Static Sinusoidal Pos<br>GANs for Image / Video<br>Human-Only RL Oversight<br>Unmonitored Latent CoT | nn.DataParallel / DDP<br>FP32 Unfused Attention<br>Dropless MoE Aux-Loss<br>Contaminated Benchmarks<br>Length-Biased RMs |
| | Scheming & Sandbag Evals<br>Aha-Moment Backtracking<br>Curriculum Difficulty RL<br>CoT Monitorability Checks | Weak-to-Strong Generalize<br>VinePPO / ReST-MCTS* Tree<br>Coconut Latent Space CoT<br>Specification Gaming Fix | Cross-Modal Contrastive<br>MMDiT Dual-Stream Blocks<br>SoundStorm Parallel Audio<br>Spatio-Temporal Patchify | Rotary 3D-RoPE Video Pos<br>VQ-GAN / FSQ Scalar Quant<br>Latent Concept Erasure<br>Distribution Match Distill | CFG Guidance Distillation<br>MAR Masked Autoregressive<br>Tree-Ring Latent Watermark<br>Physics Video Priors | ASL / CBRN Frontier Evals<br>DualPipe Compute Overlap<br>Ring-Attention 1M+ Context<br>Micro-Scaling NVFP4 | Constitutional Classifiers<br>Multi-Token Predict (MTP)<br>FlexAttention & Pallas<br>Zero-Bubble Pipe Parallel | Chinchilla / MuP Scaling<br>WSD Warmup-Stable-Decay<br>NVSHMEM All-to-All MoE<br>Differential Privacy SGD | | |